

Hospitals



Local training



Evidence



Reference Encoder

Class prototype



Diagnostic Head

Extract class-level evidence

Diagnosis prototypes



prototype



prototype



prototype

Diagnosis prevalence counts



Raw images never leave the hospital

DPR provides the **diagnostic head** w_c
LPC provides the **prevalence prior** b_c
they establish a **global diagnostic view**:

$$l_c(x) = w_c^T \phi_0(T(x)) + \mathbf{1}[r > \tau] b_c$$

LAMP-Merge

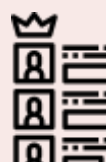


Diagnostic Prototype Reconstruction

1. Prototype Extraction

prototype
prototype
prototype

Evidence



Reallocate weights

prototype
prototype
prototype

2. Reconstruct Diagnostic Head



Head

$w_1, w_2 \dots$

Long-tail Prevalence Calibration

1. Prevalence Prior



Prior distribution

2. Long-tail Calibration



Class k is common!

$b_1, b_2 \dots$

prototypes
support

counts

Global Diagnostic Model



Reference Encoder

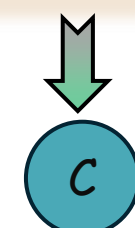
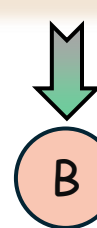
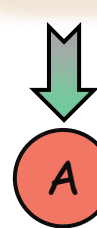
Class prototype

Diagnostic Head

$$w_c = [w_1, w_2, \dots, w_n]$$

$$b_c = [b_1, b_2, \dots, b_n]$$

Reconstruction evaluation metric



Robust prediction